

Pavan Kalyan Natukula

Npavankalyan2023@gmail.com | +1 (940)-808 8962 | <https://www.linkedin.com/in/pavankalyanatukula/>

CAREER SUMMARY

Experienced AI/ML and Data Science professional with over 3 years of hands on experience designing intelligent and data driven systems. Experienced in building scalable ML pipelines, AI and RAG applications that combine automation with actionable insights. Passionate about turning complex data in to simple, impactful solutions using python, cloud technologies, LLMs, and modern machine learning frameworks

TECHNICAL SKILLS

AI & ML: TensorFlow, Keras, Scikit-Learn, PyTorch, LlamaIndex, RAG, Langchain, LLM's, CNN's, RNN's, Hugging Face.

Data Engineering: BigQuery, Dataflow, DataStream, Pub/Sub, Cloud Composer, Apache PySpark, Apache Beam

Cloud & Big Data: AWS Cloud (S3, EMR, Lambda, Sagemaker), Docker, Kubernetes

Languages: Python (Pandas, NumPy), SQL, PL/SQL, Linux

Databases: Oracle DB 19C, Microsoft SQL Server, PostgreSQL.

Data Visualization: Tableau, PowerBI, Matplotlib, Seaborn, Plotly

Automation & Orchestration: Git, GitHub Actions, CI/CD pipelines, Airflow, N8N

PROFESSIONAL EXPERIENCE

AI Engineer Intern | DYCSI, USA

Aug 2025 – Dec 2025

- Reduced RFP proposal generation time by 95% from days to 30 minutes, targeting a 5x throughput increase, and 20% higher win rate by developing Project Archietct, a production grade RAG application that automates enterprise RFP responses.
- Deployed the containerized RAG application to google cloud using Docker Compose, orchestrating Ollama, ChromaDB, FastAPI, and Next.js in production environment.
- Implemented multimodal RAG for PDF images and tables using VLM's, improving retrieval response by 35%.

Data Engineer | Cognizant Technology Solutions, India

May 2021 - Dec 2023

- Processed 10M+ daily records with reduction in data latency (24 hours to 5 minutes) by designing end to end ETL pipelines on Google Cloud that integrated sales, marketing, financial, customer, privacy, and, pps date from 10+ source systems, enabling real time analytics and ML ready feature generation.
- Partnered with business stakeholders to analyze large scale operational data, performing EDA and statistical profiling, uncovering signals for predictive analytics and ML use cases.
- Implemented ML based failure prediction with 85% accuracy by analyzing historical execution logs, resource utilization patterns, and data quality metrics.
- Improved data pipeline reliability from 70% to 90% by engineering ETL pipelines with deduplication, data validation, and DLQ mechanisms that processed 300+ records monthly.
- Enabled data driven insights supporting \$50M+ annual event revenue by designing a three layer EDW with 35+ fact/dimension tables (SCD Type 2), delivering 20+ analytics reports for strategic planning of 200+ events.

CERTIFICATIONS

- Google Cloud Certified Professional Data Engineer

PROJECTS

Smart Tutor AI Driven Personalized Teaching Support:

- Engineered production level AI tutoring system improving student engagement by 45% with 24/7 instant academic support, with context aware responses from course materials and metadata enhanced search.
- Developed end to end RAG pipeline using LlamaIndex, implementing hierarchical semantic chunking and vector based retrieval that achieved 92% answer accuracy with proper citations and 85% retrieval precision.

Medical image disease classifier:

- Developed a multiclass medical image classification machine learning model using CNN's by preprocessing diverse X-ray, CT, and MRI datasets by achieving an accuracy of 93%.
- Deployed the model on web platform using FastAPI and Uvicorn, making them accessible for real time disease diagnosis by healthcare professionals.

EDUCATION

University of North Texas | Denton, Texas

Master of Science – Data Science

Jan 2024 - Dec 2025

GPA: 4.0